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Comparative evaluation of machine learning algorithms for land use land cover mapping in Sankhuwasabha District, Eastern Nepal

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Abstract

Accurate land use/land cover (LULC) mapping in the Himalayan region remains challenging because of its rugged terrain, steep elevation gradients, heterogeneous landscapes, and high spectral variability. This study aims to evaluate the six commonly used supervised classification algorithms in Sankhuwasabha District of Eastern Nepal using open-source Landsat and Sentinel-2A satellite imagery. The algorithms considered for evaluation are Maximum Likelihood Classification (MLC), Support Vector Machine (SVM), Parallelepiped (PP), Artificial Neural Network (ANN), Random Forest (RF), and Minimum Distance (MD). Sankhuwasabha District in eastern Nepal was chosen as the evaluation site due to its complex terrain (elevation range 247–8485 m) where field visit was conducted in the north–south and east–west transect to collect the training and test samples (1100 sample points). Additional 207 ground-truth data were added from the high-resolution Google Earth Imagery of the year 2023. The images were classified into water body, cultivated area, vegetation, snow/ice, barren land and built up area. For the accuracy assessment of the classified image and evaluation of the algorithms, producer, user and overall accuracy, and kappa coefficient was used. Results show that RF provides the highest overall accuracy for both the Sentinel 2A (88.73%, Kappa = 0.81) and Landsat (84.12%, Kappa = 0.73) followed by SVM (72.73%) for Sentinel and ANN (72.27%) for Landsat, while PP provides the lowest overall accuracy value for the both satellite data. Although, RF algorithms better fits for the complex mountain landscapes, the appropriate algorithm is sensor-algorithm dependent rather than the resolution of image alone. By systematically comparing multiple machine learning and conventional classifiers across two widely used satellite sensors in a highly heterogeneous Himalayan landscape, this study provides guidance for selecting appropriate classification algorithms for accurate LULC mapping in mountainous regions.

Keywords LULC, Remote sensing, Machine learning, Mountains

1 Introduction

Land use land cover (LULC) refers to changes in the biophysical cover and use of land for different purposes [1]. LULC is a multidimensional process with a range of factors [2]. Land use relates to human activities on a piece of land, whereas land cover refers to



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the surface features [3]. LULC are critical for preserving and managing natural resources on Earth's surface and is typically caused by a complex set of economic, demographic, social, cultural, technical, and environmental variables [4]. LULC changes play a pivotal role in environmental and ecological changes [5]. The changes in land cover have many impacts on the ecosystem process, watershed area and also in the status of soil erosion [6, 7].

Remote sensing (RS) is a very useful and advance tool for assessing and examining the distribution of land, and used in comprehensive analysis of LULC [8]. RS is also used for image classification for which various factors like quality of imagery, a precise classification process and users experience, and expertise of procedure should be considered [9]. Land-use change analysis, land-use planning, environmental impact assessments, and quantitative evaluations of the risk of environmental hazards depend on the classification of LULC [10]. Supervised and unsupervised algorithms of image processing can be used for analysis of raw image data [11]. The utilization of image processing technique to classify raw satellite image provides precise information about the landscapes [12].

There are several available machine learning (ML) algorithms for image categorization. Learning techniques are divided into two categories: supervised and unsupervised techniques. Support vector machine (SVM), random forest (RF), spectral angle mapper (SAM), fuzzy adaptive resonance theory-supervised predictive mapping (Fuzzy ARTMAP), Minimum Distance (MD), radial basis function (RBF), decision tree (DT), multilayer perception (MLP), naive Bayes (NB), maximum likelihood classifier (MLC), and fuzzy logic are examples of supervised classification techniques [13], while unsupervised classification techniques include Affinity Propagation (AP) cluster algorithm, fuzzy c-means algorithms, K-means algorithm, ISODATA (iterative self-organizing data) etc. [14]. The selection of the method is influenced by factors including the site's characteristics, the data already available, and the classes' spectral similarity [15, 16]. Accurate image categorization, regardless of method, is also important [17]. More sophisticated machine learning (ML) approaches have gained popularity in recent years for LULC classification [18]. Quantitative assessment of changes in LULC is one of the most efficient ways to understand and manage land transformation, and hence there is a need to compare the accuracy of different LULC mapping algorithms to identify the best classifier for future earth observation applications [19].

Nepal has a very complex terrain with several LULC types occurring in a very small spatial extent. This demands the selection of the best classification techniques to generate accurate LULC maps. The LULC in Nepal have been constantly changing over the last few decades because of major changes induced by anthropogenic and natural forces, as well as their effects on the national and regional ecology and climate [20]. LULC change is one of the most significant types of environmental change occurring in many mountain regions across the world [21]. Machine learning-based Land-cover Classification has become much easier to access in recent years, due to the development of cloud-based platforms like Google Earth Engine (GEE) and open-source remote sensing software.

The land cover mapping in the Himalayan region has some special challenges due to topographic shadows, steep elevation gradient, rugged topography, seasonal snow cover, cloud contamination and spectral similarity among the land cover classes. These conditions bring a lot of spectral variation and make the classification even more uncertain,

especially in complex mountain environments like the Arun Valley. Therefore, good machine learning algorithms that can handle the nonlinear relationship and heterogeneous landscapes have become more crucial for better classification performance.

Although several studies have applied machine learning algorithms for LULC classification, comparatively few have systematically evaluated the performance of multiple supervised classification algorithms using both Landsat 8 and Sentinel-2 imagery under identical conditions in the complex mountainous environment of eastern Nepal [10, 22, 23]. Furthermore, the influence of sensor characteristics on classifier performance in heterogeneous Himalayan landscapes remains insufficiently explored. This study addresses these gaps by providing a comprehensive comparison of six widely used supervised classification algorithms using two freely available satellite sensors and by evaluating their classification performance in terms of accuracy, agreement, and spatial uncertainty. The findings provide practical guidance for selecting appropriate classifiers for LULC mapping in mountainous regions.

The objective of this study is to evaluate different types of image classification algorithms for land use land cover modelling in the complex mountainous terrain of Nepal. The specific objectives are to i) generate LULC model of mountain landscape using the supervised classification algorithms, ii) to evaluate the performance of the algorithms by computing accuracy matrices, and evaluate the performance of open access image (Landsat and Sentinel).

2 Study area

For this study, Sankhuwasabha District was chosen as the study site which lies in the Eastern Nepal in the Koshi Province (Fig. 1). The district covers an area of 3480 sq km and lies at Latitude 26°06" and longitude 86°57" to Latitude 27°55" and Longitude 87°40". Khandbari is the headquarter of this district. This district was chosen as the site because of its complex terrain and heterogenous landscape that this study demands. The study

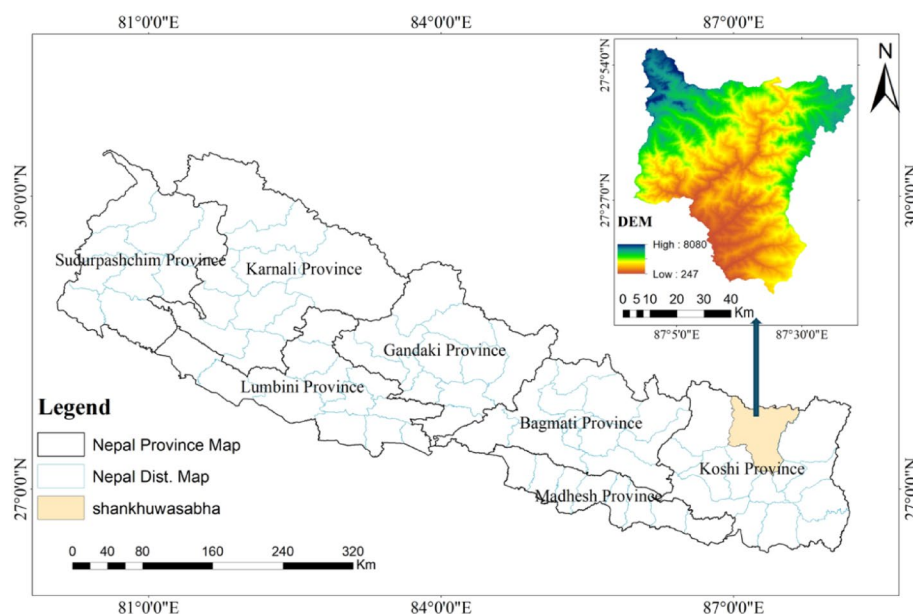


Fig. 1 Map of study area showing Sankhuwasabha District with elevation ranging from 247 to 8080 m on Nepal map

area has five climatic zone- Alpine (above 4000 m), Sub-Alpine (00–000 m), Temperate (00–00 m), Sub-tropical (00–00 m) and ropical (below 1000 m). In just a vertical span of 91 km the elevation ranges from 247 m at the deepest valley of the world, the Arun Valley up to 8485 m at the peak of the fifth highest mountain in the world. The study sites include all types of LULC snow/ice, glaciers, forest, grassland, cropland, urban, river, lakes, bare land, etc. Agriculture is the main economy of this district with crops like maize, rice, millet, sugarcane and potato as major food and cash crops.

3 Material and methods

3.1 Data acquisition and preprocessing

In this assessment, satellite imagery from two data sources were considered. Sentinel 2A imagery of 10 m and Landsat 8 imagery of 30 m were used. Landsat 8 data was atmospherically corrected to surface reflectance using FLAASH and Sentinel 2 data was atmospherically corrected using Sen2Cor (SNAP). The image was not processed with any additional topographic correction (C-Correction, Minnaert or SCS+C) before classification. The surface reflectance products were thus corrected for atmospheric effects and the classification algorithms were trained with these products.

3.1.1 Sentinel 2A

Sentinel 2A imagery data was downloaded from the Copernicus Scientific Data Hub website (<https://sentinels.copernicus.eu/web/sentinel/user-guides/sentinel-2-msi>). The two tiles of the sentinel-2A covered the whole study area which were mosaiced together. Sentinel Application Platform (SNAP) was used for the atmospheric correction of the image. Then, the 10 m bands (2, 3, 4, and 8) were stacked together to form a multiband imagery. The detail of data acquired is listed in Table 1.

3.1.2 Landsat-8

Landsat-8 OLI imagery data was freely downloaded from the United States Geological Survey (USGS) Earth Explorer website (<https://glovis.usgs.gov/>). The whole study area was covered in a single tile of Landsat-8 image. The detail of data acquired is listed in the Table 1. The pre-processing of landsat-8 image involves the topographic correction, radiometric calibration, top of the atmosphere reflectance and surface reflectance computation which was done in the ENVI 5.3 software. offers various radiometric correction methods such as Dark Object Subtraction (DOS), Fast Line-of-sight Atmospheric Analysis of Spectral Hypercubes (FLAASH), Empirical Line and Flat Field. FLAASH method was used for the radiometric correction of the landsat-8 image. The flowchart of methodology for this research is presented in the Fig. 2.

Table 1 List of selected Sentinel 2A and Landsat 8 images for the study area

Data source	Granule ID	Sensing date	UTM zone	Cloud percentage
Sentinel	L2A_TL_2APS_20231203T081149_A0441117_T45R	12/3/2023	45	<5
Sentinel	L2A_TL_2APS_20231203T081149_A0441117_T45R	12/3/2023	45	<5
Landsat	LC08_L1TP_140041_20231123_20231128_02_T1	12/3/2023	45	<5

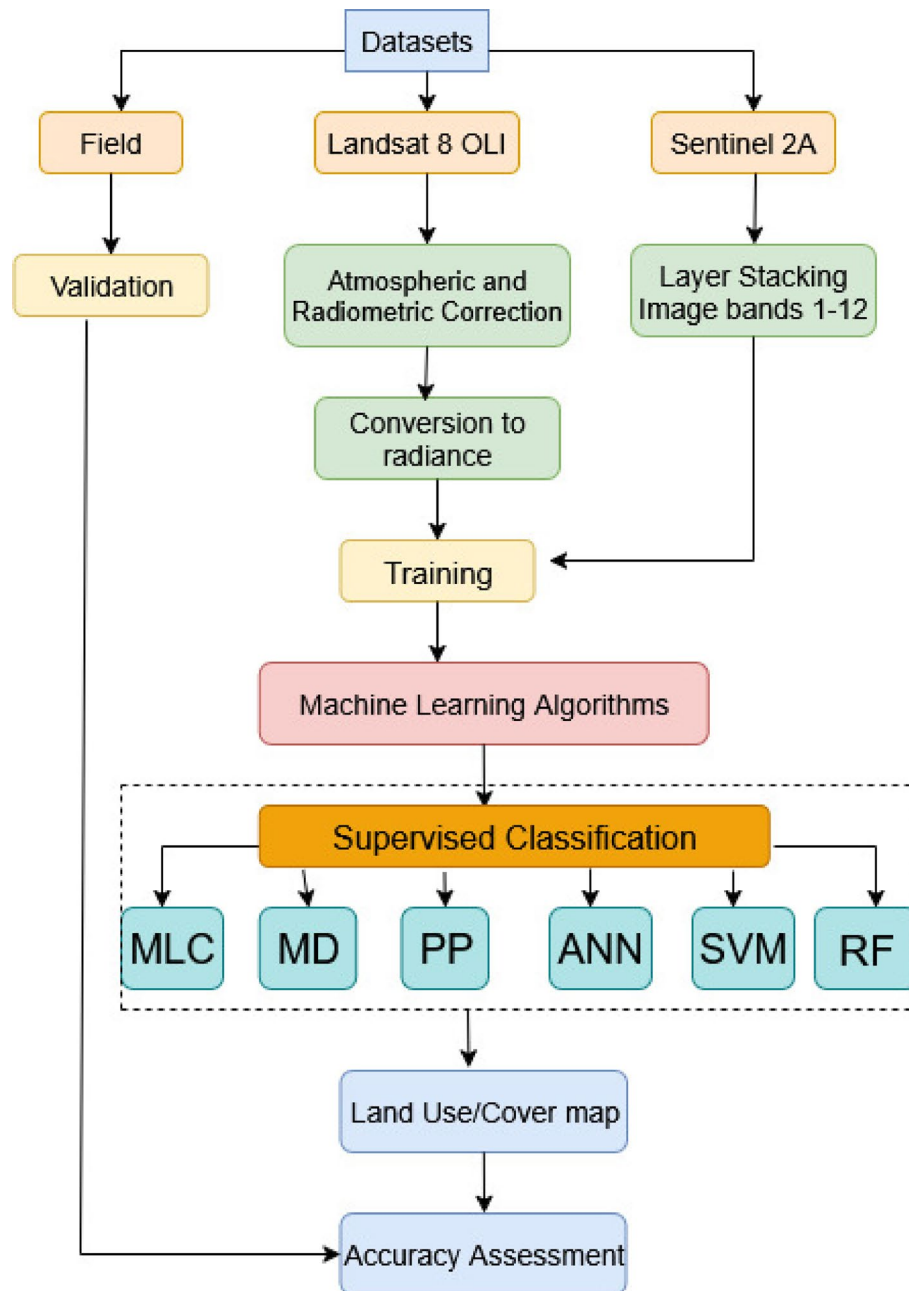


Fig. 2 Methodological framework used in this study

3.2 Training and testing sample collection

Training and testing data sets were identified for the different land cover types by using stratified random sampling based on preliminary image screening and field knowledge. These land use land cover types include water body, built up area, vegetation, cultivated area, snow/ice and barren land. In the field, a north-south and east-west transect survey was done using Global Positioning System (GPS) device to collect the random ground truth samples with adequate representations of each classes. The GPS points from the field and high-resolution orthorectified Google Earth imagery was used as the reference for the selection of the training and test samples. The total number of samples collected was 1370 (1100 from fields and 270 from google earth image), out of which about 35%

were used for the accuracy assessment of the models. A 30 m × 30 m grid was established across the study area to spatially distribute the samples, ensuring that sampling points were at least 50 m apart. This helped minimize spatial autocorrelation and improve the spatial independence of the samples. The number of samples for each class is shown in Table 2.

3.3 Image classification algorithms

Six image classification algorithms (Maximum likelihood classification, support vector machine, Parallelepiped, minimum distance, random forest and Artificial neural network) were used and evaluated in this study.

3.3.1 Maximum likelihood classification

One of the most well-known and often employed classification algorithms in remote sensing is maximum likelihood classification (MLC), which is regarded as a fundamental pixel approach [24]. This method relies on the basis of that each class's data from each band has a normal distribution [25]. In order to obtain an accurate estimate of the mean vector and the variance–covariance matrix, only a small number of pixels need to be chosen and for the method to account for the numerous variations in spectral features, the greatest number of samples should be used. The following Eq. 1 is the computation of this likelihood using Bayesian law [26]:

$$D = \ln(a_c) - [0.5 \ln(|COV_c|)] - [0.5 (X - M_c)^T (COV_c^{-1}) (X - M_c)] \quad (1)$$

where c is the required class, X is the desired pixel's measurement vector, M_c is the class c mean vector, and COV_c is the covariance matrix of the class c pixels. D (weighted distance) denotes the probability in this equation.

3.3.2 Support vector machines

A binary based on optimum separating hyperplane and statistical machine learning theory is used in the Support Vector Machine (SVM) model, a high-performing supervised machine learning approach [27, 28]. To adequately distinguish between the presence and absence classes of used samples for a maximum of the classification margin in a feature space is the goal of the optimal classification hyperplane.

In general, there are two types of SVM models: two-class (mostly used model) and multi-class (which combine several two-class SVMs) [29]. The separation hyperplane (H) is the likely plane for dividing the two classes throughout the SVM procedure by utilizing the following Eq. 2 [27]:

Table 2 The training and testing samples for land use/cover classification

Class	Training sample	Validation sample
Water body	155	50
Cultivated area	106	40
Vegetation	150	45
Snow/Ice	169	55
Barren land	339	120
Built area	101	40
Total	1020	350

$$\text{Min } \omega, b, \xi : \frac{1}{2} \omega^T \omega + c \sum_{i=1}^1 \xi_i \quad (2)$$

where w , b , ξ_i , and c (>0) stand for a coefficient vector, the starting point offset of the hyperplane, the positive slack variable, and the error penalty parameters, respectively.

3.3.3 Parallelepiped classification

Parallelepiped (PP) classification uses a simple decision rule to classify hyperspectral data. The decision boundaries form an n -dimensional parallelepiped in an image data space. The dimensions of a PP classifier are defined based on the standard deviation threshold from the mean of each selected class [30, 31]. Pixels are classified if the threshold range falls within the specified low and high values. Those area were classified as unclassified area that do not fall into the given threshold range [11].

3.3.4 Minimum distance

The minimal distance (MD) classification approach is based on measuring the distance between unknown pixels and each class's mean vectors. In other words, the pixel is given to the class whose mean values are closest to the desired pixel [32, 33]. Its simplicity in mathematics and the requirement for a mean vector for every band in the training data made this method popular. In this method, operation is quickly since it doesn't require complicated calculations because of its straight forward logic. Furthermore, this method is not appropriate in regions where spectral classes are extremely close to one another, according to following Eq. 3 [34]:

$$\text{Dist} = \sqrt{(BV_{ijk} - \mu_{ck})^2 + (BV_{ijl} - \mu_{cl})^2} \quad (3)$$

where Dist (distance) is the distance of mean score to the unknown pixel), μ_{ck} is the mean vector of the class c in the band k , and μ_{cl} is the mean vector of the class c in the band l .

3.3.5 Artificial neural network

A nonlinear modeling tool called an Artificial Neural Network (ANN), which is based on the biological neural network of a human, can answer issues devoid of any presumptions. It may therefore find intricate connections between different input data types [35, 36]. The general structure of this method consists of three different layers. In the first layer, data is received. The required computations are made based on the neurons (processing units) on the data in the second layer, sometimes referred to as the hidden layer. In the last layer, final outputs were specifying. Through a non-linear optimization process (training), weights and biases are identified with the goal of minimizing a learning function that expresses the degree of similarity between the observation and the output of the ANN [18].

Let n input neurons be represented by $u = (u_1, u_2, \dots, u_n)$ and output neurons by $v = (v_1, v_2)$. The activation function used in hidden neurons for classification is represented as Eq. 4:

$$v = f \left(\sum_{i=1}^n w_i u_i + \beta \right) \quad (4)$$

where v and β represent the bias and ω_{ji} are the linked weights between input neurons u_i and output neurons.

3.3.6 Random forest

Random forest (RF) is a non-parametric machine-learning algorithm which has been widely applied for solving the environmental problems, like water resource management and natural hazard management [37]. It can handle a variety of data, like satellite imagery, and numerical data [38]. It is an ensemble learning method based on a decision tree, which combines with massive ensemble regression and classification trees. Two parameters, (i) the number of trees, which can be explained by 'n-tree' and (ii) many features in each split, which can be explained by 'm-try' are needed for setting up the RF model. Classification trees provide an individual choosing power or vote and accurate classification in regulating the majority vote from trees in the entire forest [37].

A RF consists of a combination of classifiers where each classifier contributes with a single vote for the assignation of the most frequent class to the input vector (x),

$$\hat{C}_{rf}^B = \text{majority vote} \left\{ \hat{C}_b(X) \right\}_1^B \quad (5)$$

where $\hat{C}_b(x)$ is the class prediction of the random forest tree [65].

RF is an ensemble classification algorithm which uses trees as base classifiers, $\{h(x, \Theta_k), k=1, \dots, B\}$, where x is the input vector and $\{\Theta_k\}$ are the independent and identically distributed random vectors [66].

The parameters used to run different models is given in Table 3.

3.4 Data analysis

Environment for Visualizing Image (ENVI 5.3) software was used to generate the LULC model by using the MLC, MD, PP, ANN, and SVM algorithms and R software was used to generate the LULC model using the RF algorithm.

Confusion matrix was generated in the QGIS with the help of accuracy/ validation point of from the field. The confusion matrix allows calculating the overall accuracy (OA) and kappa statistics (K) and shows where the classification generates confusion (omission and commission errors, quantified separately by the user's accuracy and producer's accuracy) among the LULC classes producing inaccuracies [39]. The statistics was used to check and compare the performance of the algorithms [18, 40].

Table 3 Model parameterization

Algorithm	Parameterization
SVM	RBF kernel, Penalty parameter (C) = 10, $\gamma = 0.1$
Random Forest	300 trees, variable per splits = 2.65
Neural Network	Network type: Multilayer Perceptron (MLP), 1 hidden layer, 10 neurons, Learning Rate = 0.01
MLC	Equal priors
Minimum Distance	Euclidean distance
Parallelepiped	Mean $\pm$ 1.5 SD

3.4.1 User and producer's accuracy

The user's accuracy relates to the possibility of a pixel belonging to a given class on the map. The calculation (Eq. 6) involves dividing the number of correctly classified pixels by the total number of pixels in the category [41].

$$User\ accuracy\ \% = \frac{Number\ of\ correct\ samples\ of\ a\ class}{Total\ samples\ in\ row} \times 100 \quad (6)$$

The producer's accuracy refers to the chance of a specific ground attribute being classified as such. It is computed (Eq. 7) by multiplying the number of pixels accurately categorized in each category by the number of pixels sampled for that category [41].

$$Producer\ accuracy\ \% = \frac{Number\ of\ correct\ samples\ of\ a\ class}{Total\ samples\ in\ column} \times 100 \quad (7)$$

Overall accuracy measures categorization accuracy. This value is calculated (Eq. 8) by dividing the number of successfully categorized pixels by the total number of reference pixels.

$$Overall\ accuracy\ \% = \frac{Total\ number\ of\ correct\ samples}{Total\ samples} \times 100 \quad (8)$$

3.4.2 Kappa coefficient

Cohen's kappa coefficient (KC) is a discrete multivariate accuracy metric. Randomly assigning pixels to classes in categorization results in a percentage accuracy value. While pixels are not assigned at random during classification, statistical techniques can account for random chance's role in evaluating classification accuracy. The Kappa measure (Eq. 9) compares classification accuracy to the likelihood of randomly assigning pixels to the correct categories, accounting for chance agreement [41].

$$K = \frac{N \sum_{i=1}^n m_{i,i} - \sum_{i=1}^n (G_i, G_i)}{N^2 - \sum_{i=1}^n (G_i G_i)} \quad (9)$$

where: i is the class number, N is the total number of classified values compared to truth values, $m_{i,i}$ is the number of values belonging to the truth class i that have also been classified as class i (i.e., values found along the diagonal of the confusion matrix) and C_i is the total number of predicted values belonging to class i , G_i is the total number of truth values belonging to class i .

3.4.3 Spatial uncertainty analysis

To investigate the spatial variability among the six classification algorithms (RF, ANN, MD, MLC, PP, and SVM), a pixel-wise standard deviation map was generated using the Cell Statistics tool in ArcGIS 10.8. The classified land-cover maps from the six algorithms were used as inputs, and the Standard Deviation (STD) statistic was calculated for each pixel. Areas with lower standard deviation indicate higher agreement among the classifiers, whereas higher standard deviation values indicate greater disagreement and spatial uncertainty in land-cover classification. The resulting standard deviation map was used as a spatial diagnostic to identify regions where model predictions were inconsistent.

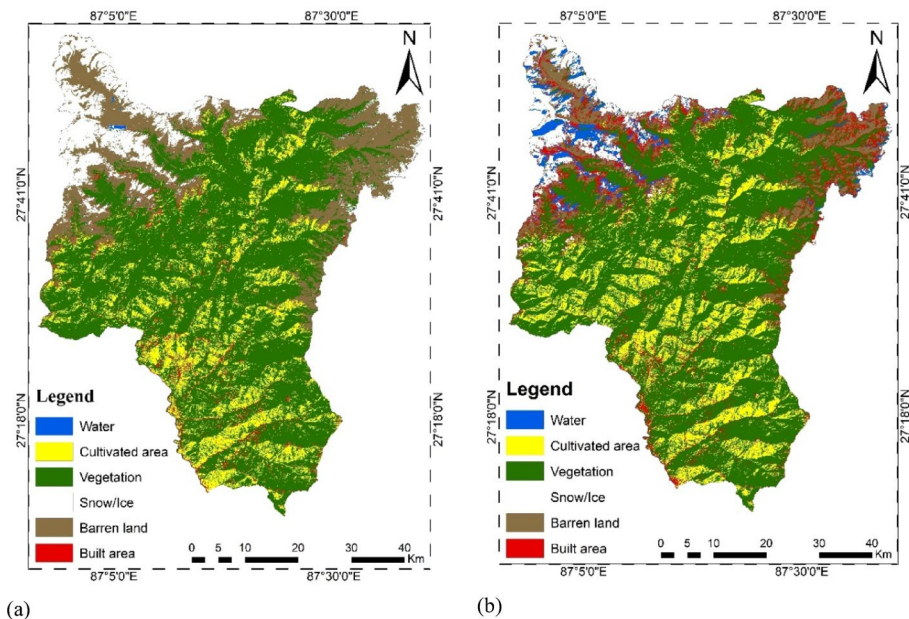


Fig. 3 LULC map generated from Landsat 8 image using different algorithms: **a** MLC, **b** MD, **c** PP, **d** ANN, **e** SVM and **f** RF

4 Results

4.1 Land use land cover models from Landsat 8

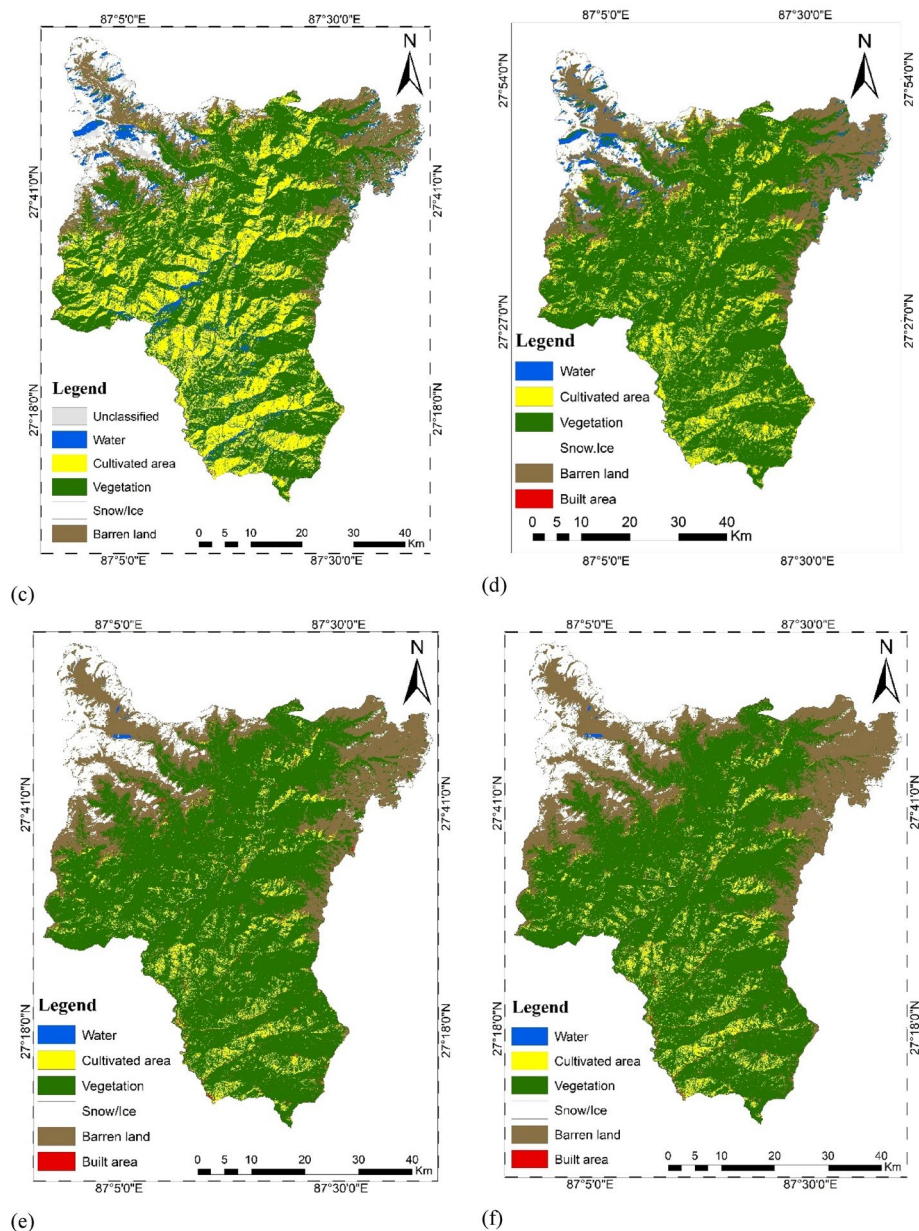
The satellite imagery was classified into six classes. This includes water body, cultivated area, vegetation, snow/ice, barren land and built area. Figure 3 shows the LULC models from Landsat 8 OLI image generated using (a) MLC, (b) MD, (c) PP, (d) ANN, (e) SVM and (f) RF. The LULC models generated by using MLC, ANN, SVM and RF clearly differentiate the LULC into six different classes. The map generated by PP model did not classify built up area adequately. The LULC model generated by using the SVM, ANN and RF model provide a better visual representation of land use than the PP and MD.

4.2 Land use land cover models from Sentinel 2A

The LULC models generated by using Sentinel 2A image were also classified into six classes. Figure 4 shows the LULC models generated by using the MLC (a), MD (b), PP (c), ANN (d), SVM (e) and RF (f). The LULC models generated by MLC, ANN, SVM and RF model clearly differentiate the six different classes. The LULC model generated by using the ANN, SVM and RF model provides the better visual representation than the PP and MD model as in the case of Landsat 8 imagery.

4.3 Accuracy assessment and evaluation of algorithms

By using the confusion matrix, the user accuracy and producer accuracy of different land use classes were assessed. The summarized accuracy assessment statistics for all algorithms for Landsat data and Sentinel data are given in Supplementary Tables S1 and S2 respectively. The higher the user accuracy, the greater the algorithm's capacity to differentiate spectrums for the appropriate class. The overall accuracy for Landsat 8 image classification using different algorithms followed the order from high to low as RF (84.12%), ANN (72.27%), SVM (71.94%), MLC (64.36%), MD (63.62%), and PP (60.85%). The Kappa Coefficient value range from 0.49 to 0.81. In case of Sentinel image,

**Fig. 3** (continued)

the accuracy of different algorithms followed the order from high to low as RF (88.73%), SVM (72.73%), MD (70.85%), ANN (70.45%), MLC (42.86%), and PP (69.22%). The detail calculation of confusion matrix for each algorithm is given in the Supplementary material (Table S3–S14).

The accuracy assessments indicate that, RF provides better accuracy in producing LULC models for both Landsat and Sentinel image followed by ANN for Landsat 8 and SVM model for Sentinel 2A data. Figure 5 shows the overall accuracy for each algorithm and comparison between Sentinel 2A and Landsat 8 data. The PP produced less accuracy for LULC models for both Sentinel 2A and Landsat 8 images.

For the Landsat 8 image (Fig. 6), RF is the best-performing model with the highest OA (84.12%) and highest Kappa (0.73). ANN (72.27%) and SVM (71.94%) form a second tier

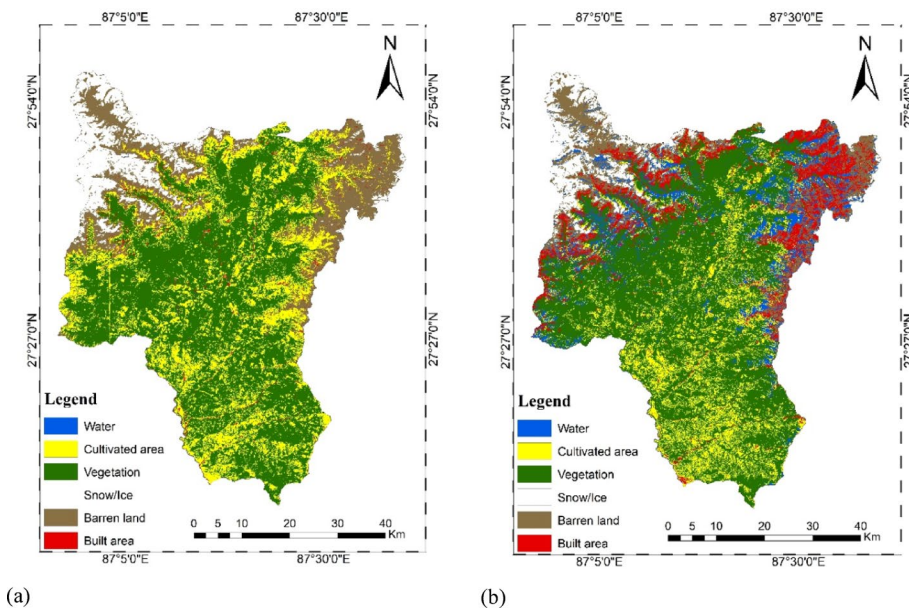


Fig. 4 LULC map generated from Sentinel 2A image using different algorithms: **a** MLC, **b** MD, **c** PP, **d** ANN, **e** SVM and **f** RF

of performance, showing moderate-to-high classification reliability while MLC (64.36%), MD (63.62%), and PP (60.85%) show comparatively weaker performance.

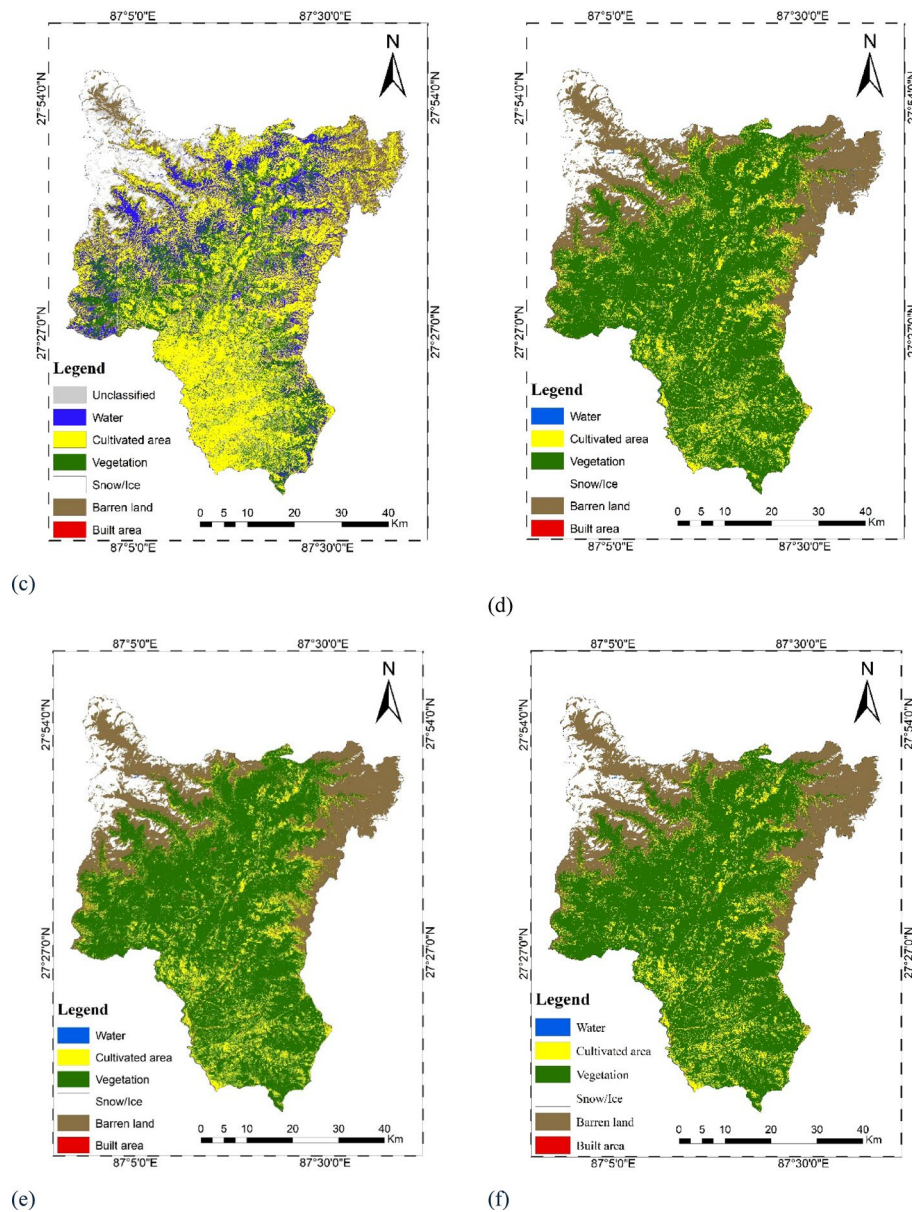
For the Landsat-8 dataset, all algorithms achieved a producer's accuracy of 100% for snow/ice and water classes. ANN, SVM, and RF showed higher producer's accuracy for vegetation, indicating better vegetation detection. In contrast, PP and ANN did not classify built-up areas effectively. Similarly, all algorithms achieved a user's accuracy of 100% for snow/ice and water, while vegetation was identified with user's accuracy ranging from 80 to 90%.

For the Sentinel 2A image (Fig. 7), RF shows clearly superior performance (OA = 88.73%, Kappa = 0.81), indicating almost perfect agreement. The SVM (72.73%), ANN (70.45%), MD (70.85%), and MLC (69.22%) show moderate performance.

Whereas PP performs very poorly on Sentinel-2A (OA = 42.86%, Kappa = 0.25), indicating poor generalization.

For the Sentinel-2A dataset, all algorithms achieved a producer's accuracy of 100% for the snow/ice class, while producer's accuracy for water ranged between 60 and 80%. RF showed the highest producer's accuracy for vegetation, whereas other algorithms achieved values below 80%. ANN recorded a relatively high producer's accuracy for barren land (about 80%), while the PP model showed the lowest producer's accuracy for this class (around 35%). In contrast, the PP model achieved a high producer's accuracy for cultivated areas (approximately 90%). In terms of user's accuracy, all models achieved values ranging from 80 to 90% for the snow/ice class.

Result shows that Sentinel-2A generally shows higher OA and KC compared to the Landsat 8, but statistical tests showed this improvement is not significant. A Z-test comparing Kappa coefficients between sensors [42–44] confirmed that the accuracy improvement of Sentinel-2A over Landsat 8 was not statistically significant for five of the six classifiers (MLC: $Z = -0.97$, $p = 0.33$; MD: $Z = -1.88$, $p = 0.06$; SVM: $Z = -0.18$, $p = 0.86$; ANN: $Z = 0.35$, $p = 0.73$; RF: $Z = -1.27$, $p = 0.20$). The one exception was PP,

**Fig. 4** (continued)

which showed a significant difference between sensors ($Z = 3.84$, $p < 0.001$), but in the reverse direction, Landsat 8 outperformed Sentinel-2A for this classifier. This pattern reinforces that sensor advantage in this study is algorithm-dependent rather than a general property of spatial resolution. Nevertheless, Sentinel-2A demonstrated clear practical advantages for specific land-cover classes, particularly vegetation and cultivated land, when combined with advanced machine-learning classifiers.

4.4 Spatial uncertainty among classification models

The pixel-wise standard deviation map (Fig. 8) revealed that the six classification algorithms exhibited high agreement across most homogeneous land-cover areas, particularly in extensive vegetation and water bodies, where standard deviation values were low. Higher standard deviation values were primarily observed along land-cover transition

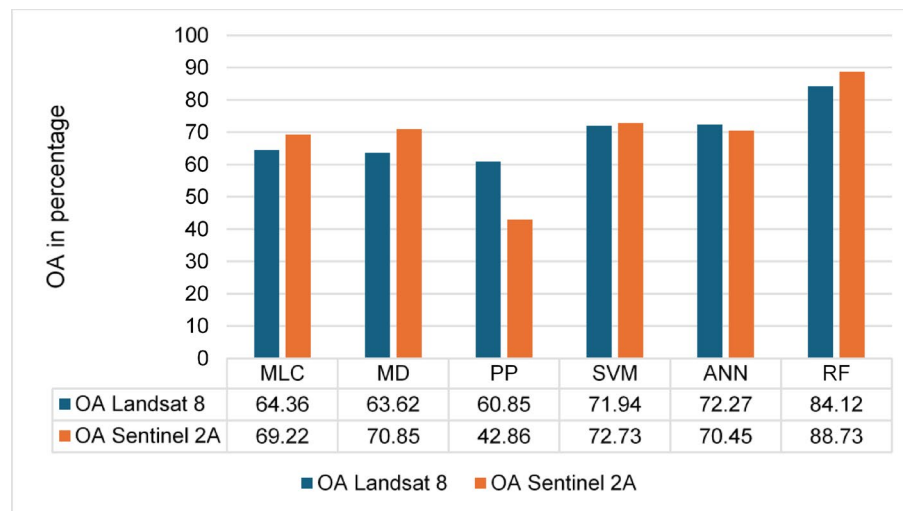


Fig. 5 Overall accuracy (OA) obtained for algorithms for Landsat and Sentinel

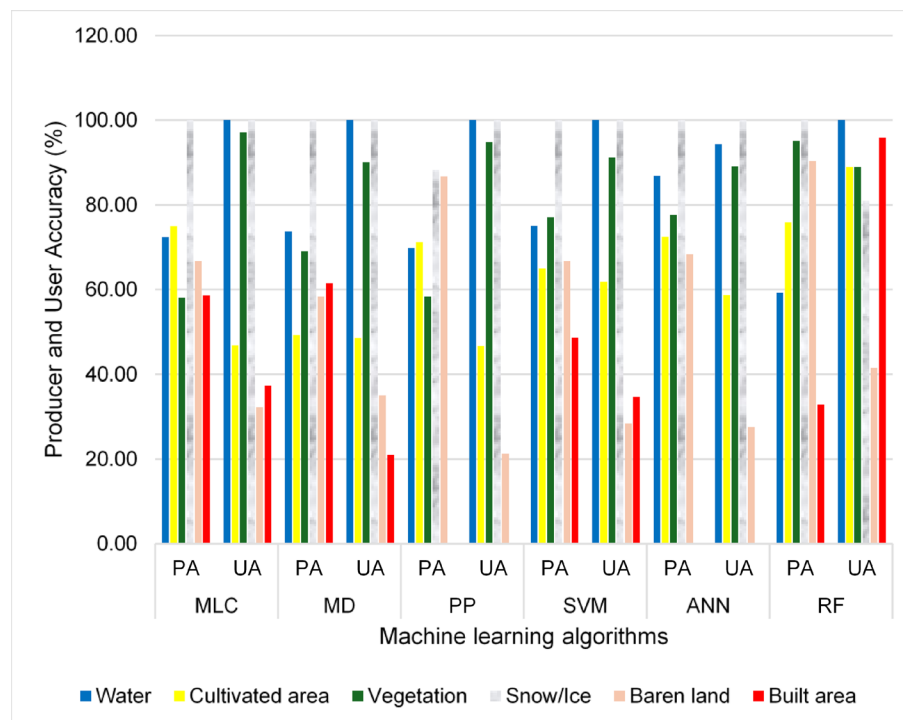


Fig. 6 Producer's and user's accuracy of Landsat 8 image for different algorithms

zones, including vegetation–cultivated interfaces, snow/ice–barren land boundaries, and topographically complex areas. These regions represent locations where the classifiers produced inconsistent predictions, indicating increased classification uncertainty.

5 Discussions

The strength of this study lies in the systematic comparison of six supervised classification algorithms across two widely used satellite sensors within a highly heterogeneous Himalayan landscape. Unlike previous studies that primarily focused on evaluating individual classifiers or a single satellite dataset, this study provides a comparative

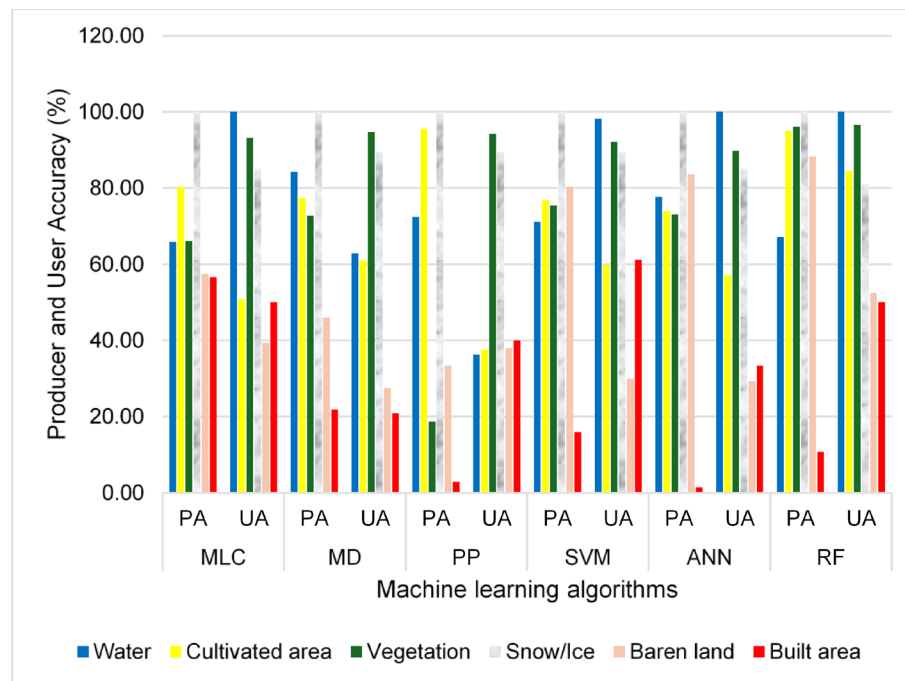


Fig. 7 Producer's and user's accuracy of Sentinel 2A image for different algorithms

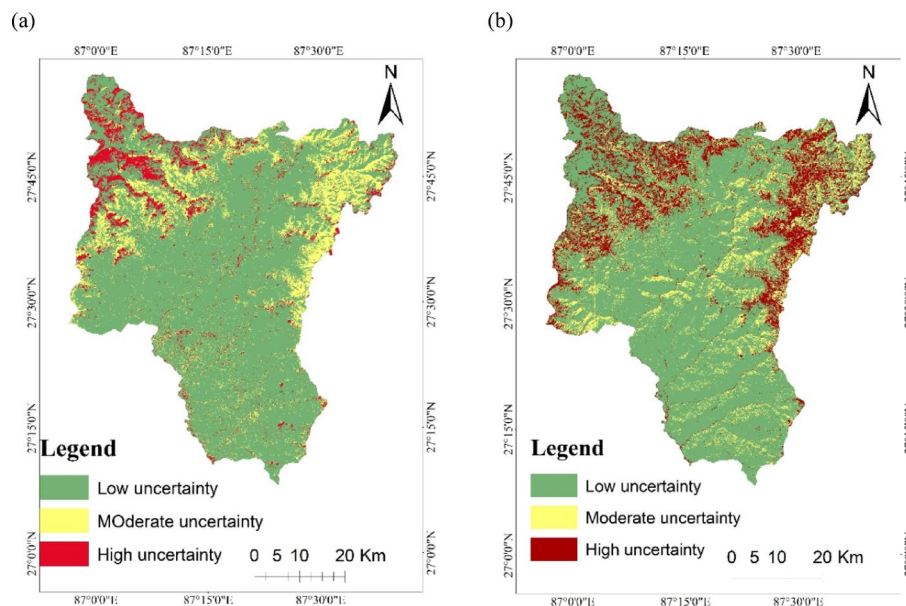


Fig. 8 Pixel-wise standard deviation map: **a** Landsat image and **b** Sentinel-2A image generated to illustrate the spatial disagreement among the six land-cover classification algorithms. Lower values indicate stronger agreement among the classifiers, whereas green color values indicate stronger agreement among the classifiers, whereas red color values indicate greater spatial uncertainty

assessment of sensor–algorithm interactions and identifies the most suitable classification approach for complex mountain environments. The inclusion of spatial uncertainty analysis further improves the interpretation of classification performance by identifying areas where classifier disagreement is greatest.

Rugged mountainous surface features are present in the study area, which can affect the surface reflectivity even after atmospheric correction due to slope, aspect and terrain shadows. As such, there could be a spectral difference between the sunlit and shaded slopes that might contribute to the spectral variability and add to the risk of misclassification, especially for spectrally similar land-cover classes like forest, barren land and shrubland. No topographic normalization was done explicitly but the use of multi-band spectral data and machine learning classifiers contributed to minimizing, though not removing, the classification uncertainty due to terrain information. Further research is warranted for using topographic correction methods, e.g., the C-Correction, Minnaert or SCS+C to enhance the performance of the classification in the mountainous area.

It should be noted that there is no explicit cross-sensor normalization performed between Landsat 8 and Sentinel-2A after atmospheric correction. Hence there can be variations in classification performance due to slight differences in reflectance characteristics arising from the use of different atmospheric correction algorithms. However, both data sets were processed to surface reflectance data using methods that are established for the respective sensors and that have been well accepted in the remote sensing community, while at the same time atmospheric effects were minimized.

Among the evaluated classifiers, RF demonstrated superior performance for Landsat-8 imagery, achieving the highest OA (84.12%) and KC (0.73). While Snow/Ice and Water classes were consistently well classified across all models, Built-up and Barren land classes exhibited significant confusion, particularly for parametric classifiers. Machine learning approaches, especially RF, substantially improved vegetation and cultivated land classification, highlighting the importance of ensemble methods for medium-resolution satellite data [45–47].

The Sentinel-2A dataset demonstrated improved classification performance compared to Landsat-8, particularly when used with advanced machine-learning classifiers. RF achieved the highest accuracy (OA = 88.73%) and almost perfect agreement (K = 0.81), with substantial improvements observed for vegetation and cultivated land classes [18, 48]. However, simpler probabilistic classifiers showed limited performance.

The spatial resolution of satellite imagery plays a significant role in land-cover classification accuracy, particularly in mountainous regions such as Nepal where land-cover types occur in small and heterogeneous patches. In many rural settlements, buildings are constructed using stone, slate, concrete, and corrugated metal roofs, which often exhibit spectral characteristics similar to the surrounding barren land and exposed rock surfaces. Consequently, Landsat imagery with a spatial resolution of 30 m is more susceptible to the mixed pixel problem, where a single pixel may contain multiple land-cover types such as built-up areas, barren land, vegetation, and cultivated fields. This spectral mixing increases the likelihood of misclassification, particularly at settlement boundaries. In contrast, Sentinel-2 imagery with a 10 m spatial resolution captures finer spatial details and better separates small settlements from adjacent land-cover classes, thereby reducing spectral mixing and improving classification accuracy [62]. Furthermore, the rugged terrain and fragmented landscape of the study area increase the occurrence of mixed pixels, particularly along the valley slopes and settlement margins. The finer spatial resolution of Sentinel-2 improves the delineation of these heterogeneous land-cover patches, resulting in greater classification consistency than Landsat imagery. These results highlight that classification performance is governed more by classifier–sensor

interaction than by sensor resolution alone. The increased uncertainty observed along land-cover transition zones is likely associated with mixed pixels, spectral similarity between adjacent classes, terrain-induced shadow effects, and heterogeneous surface conditions. In contrast, homogeneous vegetation and water bodies exhibited relatively consistent classification across all algorithms.

The spatial distribution of standard deviation indicates that topographic complexity contributed substantially to the observed classification uncertainty. Steep slopes, terrain shadows, varying illumination conditions, and abrupt elevation changes modify the spectral response of surface materials, resulting in inconsistent class assignments among different classification algorithms. Likewise, transitional zones between snow, barren land, and vegetation exhibited greater disagreement because these classes often share similar spectral characteristics under varying illumination conditions. Therefore, the spatial uncertainty analysis suggests that misclassification was influenced not only by classifier performance but also by terrain-induced spectral variability.

Recent studies from Nepal and other mountainous regions consistently show that ML approaches are more effective than traditional classifiers for LULC mapping, especially in complex terrain. Several studies conducted in Nepal have successfully applied RF and other ML algorithms using Landsat and Sentinel data. For example, Delalay et al. [49] used CART, MaxEnt, and RF with Sentinel-2 data to map land cover in mountainous areas of Nepal, while [50] applied RF to Sentinel-2 imagery to generate an LULC map of Pokhara Metropolitan City which showed that RF is an effective algorithms for mountain landscape. Similarly, Kafle et al. [51] used RF on Landsat data through Google Earth Engine to assess land-cover changes in the Chure region, and Basheer et al. [52] compared RF, CART, and MD methods to analyse land use land cover change using different satellite data. Across these studies, RF consistently produced the highest OA and Kappa values, which strongly supports the findings of the present study. Our results align well with other study as well such as Acharya et al. [53] and Singh and Pandey [54], who also reported RF as the most accurate classifier for Landsat-based LULC mapping, followed by SVM. This suggests that ensemble-based methods like RF are better able to handle spectral variability and mixed pixels, which are common in mountain landscapes [55, 56].

Comparative studies using both Landsat and Sentinel imagery further support our observations. Ghayour et al. [18], working in western Iran, compared Sentinel-2 and Landsat-8 data using multiple classifiers and found that advanced ML algorithms such as SVM and RF generally outperformed traditional methods. Similarly, Srivastava et al. [41] and Pokhariya et al. [57] showed that Sentinel-2 data often provides better class separability, particularly for vegetation, due to its higher spatial resolution and additional spectral bands [58]. However, these studies also noted that the performance gain from Sentinel-2 is not always uniform across classifiers, which is consistent with our finding that improvements were algorithm-dependent rather than sensor-dependent alone [59].

In our study, SVM and ANN performed reasonably well for vegetation classification, which agrees with findings from Aryal et al. [60] and Balha et al. [61], who highlighted the strength of ML models in capturing vegetation heterogeneity. At the same time, persistent confusion in built-up and barren land classes was observed for both Landsat-8 and Sentinel-2A. Similar challenges have been reported by Singh and Pandey [54] and

Talukdar et al. [19], particularly in mountainous or rugged regions where built-up areas are small, scattered, and spectrally similar to bare soil or rocky surfaces.

The low performance of classifiers such as PP and MD in our results is also supported by earlier studies. Talukdar et al. [19] reported minimum distance classifiers to have the lowest accuracy in heterogeneous landscapes, which aligns with our observation that PP and MD produced the weakest results. The limited use of PP in recent literature further suggests that such simple classifiers are less suitable for complex mountain environments.

In several studies, multiple algorithms have been combined for improving classification accuracy. Alshari et al. [62] have combined ANN and RF to produce high accuracy land use land cover model by using Sentinel 2A and Landsat 8. The accuracy was higher compared to ANN and RF as an individual classifier. Other studies suggest, deep learning to improve the classification [63, 64].

This study has several limitations that should be acknowledged. First, the analysis was based on a single study area and limited seasonal imagery, which may restrict the generalizability of the results to other geographic regions or climatic conditions. Second, the reference data used for accuracy assessment, although carefully prepared, may contain positional and thematic uncertainties, particularly for spectrally mixed classes such as built-up and barren land in complex mountainous terrain. Third, only a selected set of classification algorithms and default or commonly used parameter settings were evaluated; further tuning or inclusion of deep learning-based approaches could potentially improve class discrimination and fourth, the topographic effects which strongly influence spectral responses in mountainous landscapes, were not explicitly corrected and may have contributed to misclassification in certain land-cover classes.

6 Conclusion

This study evaluated six supervised classification algorithms, Maximum Likelihood Classification, Support Vector Machine, Parallelepiped, Minimum Distance, Artificial Neural Network, and Random Forest, using both Landsat 8 and Sentinel-2A imagery for land use land cover mapping in Sankhuwasabha District, a topographically complex Himalayan landscape. The comparative design, applied consistently across two independent sensors, allowed the study to separate the effect of the classification algorithm from the effect of the sensor, which is a distinction that few earlier studies from mountainous Nepal have addressed directly. Random Forest consistently outperformed the other five algorithms on both datasets, achieving the highest overall accuracy and Kappa coefficient for Landsat 8 (84.12%, $K = 0.73$) and for Sentinel-2A (88.73%, $K = 0.81$). Sentinel-2A produced higher classification accuracy than Landsat 8 overall, but this advantage was not statistically significant and was closely tied to which classifier was used, particularly for vegetation and cultivated land. This indicates that classification performance in mountain environments depends more on the interaction between the classifier and the sensor than on spatial resolution alone. Parallelepiped and Minimum Distance produced the weakest results on both datasets, confirming that simple parametric and distance-based classifiers are poorly suited to the spectral heterogeneity of Himalayan terrain.

The spatial uncertainty analysis added a further contribution beyond standard accuracy statistics. Agreement among classifiers was high in homogeneous areas such as extensive vegetation and water bodies, but disagreement increased sharply at land cover

transition zones, including vegetation to cultivated land boundaries and snow or ice to barren land boundaries. This shows that accuracy metrics alone can understate the practical difficulty of mapping mountain landscapes, and that transition zones deserve targeted attention in future classification efforts.

Taken together, these results make three main contributions. First, they provide direct, controlled evidence that Random Forest is the most reliable classifier for LULC mapping in complex Himalayan terrain across two of the most widely used open-access sensors, which offers practical guidance for researchers and land managers working in similar mountain environments. Second, they demonstrate that the choice of classifier matters more than the choice of sensor when working in heterogeneous mountain landscapes, which should inform how future studies prioritize their methodological decisions. Third, the spatial uncertainty mapping approach used here offers a diagnostic tool that identifies where and why classifications fail, which is more actionable for practitioners than overall accuracy figures alone. For applied purposes, an accurate LULC map generated using the methods evaluated here can support resource management for soil, vegetation, and water, and can help detect land use change over time, including deforestation, urban expansion, and land degradation. This information is directly relevant to environmental monitoring, policy making, and to urban planners making decisions about land development, zoning, and infrastructure in mountain regions.

Supplementary Information

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Supplementary Material 1

Author contributions

B. M.: Data collection, Formal analysis, Writing—original draft and R.C.: Writing—review & editing, Supervision, Resources, Methodology. All authors have approved it for publication.

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Data availability

The data collected and used in this study is clearly stated in the methodology sections. No additional datasets were generated.

Declarations

Ethics approval and consent to participate

Not Applicable. The study didn't involve human participant or animal subjects.

Consent for publication

All authors and the affiliated organization agree to publish this paper. The authors grant this journal permission to publish this article.

Competing interests

The authors declare that there is no any conflict of interest regarding the publication of this work.

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